

Environmental Boundary Spanning and Information Processing Effects on Organizational Performance¹

MARC J. DOLLINGER
University of Kentucky

The boundary spanning activity of the entrepreneur is used to examine the strategic management process in small business. In a sample of 82 owner/operators, intensive boundary spanning activity was strongly related to organizational performance, and information processing capability significantly affected the performance-boundary spanning relationship.

The strategic management of small business organizations has been a generally neglected area of research. This is the case in spite of the numerous calls for investigations into the strategic management process and suggestions concerning how these investigations might be executed (Cooper, 1978; Dandridge, 1979; DeCarlo & Lyons, 1980; Schendel & Hofer, 1978; Sexton, 1982; Susbauer, 1978; Vesper, 1978). This is especially true for the very small business, termed by Brockhaus (1982) "microbusiness." Yet the microbusiness provides almost laboratory conditions (relative to large complex organizations) for the study of strategic management. The small business organization has been theorized to have a simple goal structure (Schendel & Hofer, 1978), a clean proximate environment (Schendel & Hofer, 1978), a direct chain of command between goal formulation and implementation (Mintzberg, 1978), and it is the basic and most populous form of business organization in this society.

This research concerns strategic management in the very small firm. The focus of this study is on the strategic behavior of the owner/operator of the small business organization, the entrepreneur.

Brockhaus (1982) notes that most investigators and writers in the field would consider the owner/operator of a small business to be an entrepreneur. The owner/operator is personally at risk but, unlike a simple

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investor, undertakes to conduct the business on a day-to-day basis. The entrepreneur is Mintzberg's "commander" form of peak coordinator, performing the strategic management tasks of balancing the interests of the environment and the firm in which he is dominant (Mintzberg, 1978). Strategic management is an entrepreneurial task (Learned, Christensen, Andrews, & Guth, 1965), and the owner/operator of the small business firm therefore is the master strategist for the organization (Mintzberg & Waters, 1982).

The problem of studying the owner/operator as strategist is that strategy making and implementation may be informal, implicit, or inarticulated. If strategy is defined as "the pattern of decisions . . . that determines and reveals [the firm's] objectives, purposes or goals, [and] produces the principal policies and plans for achieving those goals . . ." (Andrews, 1980, p. 18), then perhaps by observing the pattern of the owner/operator's behavior, the firm's strategy may be revealed. However, it is conceivable that the pattern observed appears to hold little relevance for the firm or that no pattern emerges at all. In the first instance, if the pattern of the owner/operator's decisions does not seem to be relevant to the firm, then "business" strategy is not being observed. This behavior is not related to the role of owner/operator and entrepreneur. Either someone else is fulfilling the role or the firm's executive functions have been neglected. The owner/operator's behavior is not purposeless; however, the firm cannot be considered the focal point of that behavior. In the second case, if no pattern emerges, then the strategy being observed probably is ineffective because strategy is seen as the major source of cohesiveness in the firm (Vancil, 1976). However, before the attribution of ineffective strategy is made, the criteria for performance need to be explicit. Performance criteria typically are ill-defined in entrepreneurial situations once survival is not an issue. Thus the problem for the researcher in trying to relate a pattern of decisions, or activities, (strategy) to performance is imposing.

However, Mintzberg's analysis of the peak coordinator as coalition manager provides an insight as to how to operationalize strategic activities for the small business owner/operator. Mintzberg viewed the owner/operator as the mediator between the internal and external forces. This is conceptually equivalent to viewing the entrepreneur in his boundary spanning role. Boulton, Lindsay, Franklin, and Rue described the strategic planner as a boundary spanning individual. They recognized that in the course of boundary spanning activity, information is collected which has "strategic value, [and is] relevant to short term tactical managerial decisions, or [may] concern technical developments that . . . affect an organization's technical core" (1982, p. 501). In the microbusiness, the entrepreneur occupies the boundary spanning role. Therefore the strategic activities in which the owner/operators engage manifest themselves as boundary spanning activities.

The boundary spanning model and boundary spanning activity are relatively familiar. Adams (1976) listed five types of boundary spanning activity as: filtering, transacting, buffering, representing, and protecting.

Evan (1966) believed that by studying the boundary spanning activity of an organization, one could deduce its goal structure, strategy, and implementation plan. Organ (1971) viewed the boundary spanner as the "linking pin" between the organization and its environment. Wilensky (1967) called the boundary spanner the "contact man" who mediates the paradox caused by external forces demanding flexibility and internal ones requiring orderliness and efficiency. Aldrich and Herker hypothesized that "the organization's ability to cope with environmental constraints depends in part on the ability of the boundary spanning individual to achieve a compromise between organizational policy and environmental constraints. . . ." (1977, p. 221). Therefore, it may be reasonable to view the entrepreneur's boundary spanning activity as an important aspect of the implicit strategy formulation and implementation process that must exist in the small business organization. The entrepreneur's boundary spanning activity represents both the perception of the aspects of the environment for which action is necessary and the commitment of organizational resources (the owner's time and expertise).

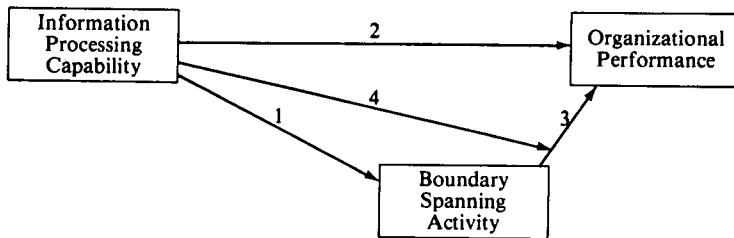
Because of the owner/operator's tremendous influence on his organization's activities, the personal characteristics of this individual are important to organizational performance. Two streams of research converge on this matter. The relationship between personal characteristics and boundary spanning has been summarized by Miles (1980). Among the correlates mentioned are flexibility, extroversion, tolerance of ambiguity, self-assurance, need for visibility, and *savoir faire*. The relationship between personal characteristics and entrepreneurship has focused on the tendency to become an owner/operator of a small business. Among the possible correlates are: need for achievement (McClelland, 1971), locus of control (Borland, 1975; Shapero, 1980), leadership and independence (Hornaday & Aboud, 1975), risk-taking (Brockhaus, 1980; Shapero, 1980), and integrative complexity (McGaffey & Christy, 1975).

Because this study examines the boundary spanning activity of the owner/operator, it would seem appropriate to concentrate on the personal characteristics of entrepreneurs who interact with the environment. The environment has been characterized in terms of complexity, change, heterogeneity, and uncertainty (Conrath, 1967; Drucker, 1980; Lawrence & Lorsch, 1967; Terreberry, 1968; Thompson, 1967). Weick (1969) advocates that the salient features of the environment are only those that are "enacted" by the individual's perception. It appears that the manner in which the individual processes information and the choice of which information to process are the important personal characteristics for the small business owner/operator in the boundary spanning role.

Figure 1 represents the general model for this research.

Arrow 1 indicates the relationship between information processing capability and boundary spanning activity. The more sensitive the individual is to salient information in the environment, the more boundary spanning activity will be performed (Kahn, Wolfe, Quinn, Snoek, & Rosenthal, 1964;

Figure 1
General Model of the Interaction of Boundary Spanning Activity, Organizational Performance, and Information Processing Capability



Leifer & Delbecq, 1978). Also, if the boundary spanning of the owner/operator is considered as a surrogate for structure (Evan, 1966), then the more complex information capabilities will require more boundary spanning. Individuals who can discriminate among a wide variety of stimuli possess a larger potential for information processing (Schroder, Driver, & Streufert, 1967). This increases their capacity for boundary spanning activity. Individuals who cannot deal with a wide variety of stimuli focus on only the least ambiguous information and have a decreased capacity for boundary spanning. By this reasoning, the entrepreneur who attends to only unambiguous events has no need for boundary spanning and strategic planning.

H1: There is a positive relationship between integrative complexity and boundary spanning activity.

H2: There is a negative relationship between intolerance of ambiguity and boundary spanning activity.

Arrow 2 indicates the relationship between information processing capability and organizational performance. Because boundary spanning is not the only managerial and entrepreneurial activity of the owner/operator, higher levels of information capacity contribute to performance—for example, through the design of more efficient production methods or control systems. In addition, the increased information processing capability results in more finely tuned strategic plans. The organization's financial performance will be better as a result.

H3: There is a positive relationship between integrative complexity and financial performance.

H4: There is a negative relationship between intolerance of ambiguity and financial performance.

Arrow 3 shows the relationship between boundary spanning and performance. Higher levels of boundary spanning are associated with higher levels of financial performance. The small business faces a turbulent environment (Child, 1972; Drucker, 1980) and requires boundary spanning activity to maintain its position and equilibrium (Benson, 1975). Entrepreneurial boundary spanning activity provides the owner/operator with the information necessary for strategy formulation. Simultaneously, the transactions between the entrepreneur and the environment implement the strategy.

The hypothesis below posits the direct link between strategy and performance.

H5: There is a positive relationship between boundary spanning activity and financial performance.

Arrow 4 indicates that the information processing capability of the owner/operator may affect the boundary spanning-performance relationship. To wit, if the capability to interpret and recode information from the environment is lacking (low capability), the boundary spanning performed would be ineffective and not contribute to the firm's financial performance. The implication is that only "good" strategy is effective. The strongest boundary spanning-performance relationships would be found in individuals with high information processing capability.

H6: When the owner possesses high integrative complexity, the boundary spanning-financial performance relationship will be stronger than that found in the low integrative complexity owner.

H7: When the owner possesses low intolerance of ambiguity, the boundary spanning-financial performance relationship will be stronger than that found in the high intolerance of ambiguity owner.

Two assumptions concerning the operationalization of the model should be made explicit. The first is that the owner/operator is the dominant boundary spanner in his organization. If boundary spanning is delegated to a considerable extent, tests of the model will be inconclusive because the wrong subject is involved. The second assumption is that the owner/operator's boundary spanning is nontrivial. That is, the behavior under examination occurs often, and it represents a major activity for the subject. These two assumptions are subject to a validity check: Is the percentage of time spent in boundary spanning relatively large in proportion to other owner/operator activities? What percentage of the organization's total boundary spanning is performed by the owner/operators? Answers to these two questions are provided in the results section.

Method

Sample

Data were collected by survey instrument from the owner/operators of four different types of small business organizations. There were 38 retail firms (food and apparel) and 44 manufacturing firms (food and apparel). Subjects were selected at random from phone directories in the Allentown-Bethlehem-Easton Pennsylvania SMSA. Of the owner/operators contacted by phone prior to mailing the survey, 180 met the inclusion criteria. The criteria were:

- firm be owner operated (≥ 51 percent ownership)
- firm have between 2 and 100 employees
- firm be in operation one year or longer

The total number of usable responses was 82 (45.5 percent). Among the demographic characteristics of the informants sampled were: median age of the owner/operator, 45 years; educational level, mode = some college; there were 66 males and 12 females (4 missing data). The average age of the firms was 29.46 years, and the average number of employees was 46.35 (mean).

A potential problem exists in the use of informants in organization research. Seidler (1974) reports that informants, people who give descriptions of relationships and patterns of behavior, may be biased. The bias increases the more vague and controversial the information and the larger the unit size being observed. Researchers sensitive to these issues—for example, Dewar and Werbel (1979)—have used multiple informants in each unit.

In this study, only the owner/operator was surveyed in each firm. However, the nature of the information—that is, to report the number of hours and contacts with people outside the organization—appears neither vague nor controversial. Seidler's (1974) work was in the context of leadership, conflict, and satisfaction in Roman Catholic dioceses and included measures of leadership style, ideological polarization, satisfaction, and conflict between bishops and clergy. These clearly are controversial issues, and biases were observed in the informants. No such biases are expected in this study.

The size of the unit within which the informant is reporting presents a different problem. Seidler's dioceses ranged from 1,400 members for large cities to 20 members for the smallest unit. His results indicate that bias (low interrater reliability) increases the larger the unit size. Even though the firms under investigation in this study are small businesses, it may be that the owner/operator is unaware of the amount of additional boundary spanning activity taking place in his organization. This poses two problems: (1) the amount of others' boundary spanning activity reported may be biased (direction unknown) and (2) the effect of the boundary spanning on performance is unmeasured.

However, these problems are not insurmountable. Seidler suggests guidelines to counteract bias when selecting informants. Among these are: (a) select a balanced set of informants (expecting bias in both directions); (b) standardize the informants by selecting occupants of the same positions; and (c) select informants who have access to the information needed. These three criteria to minimize bias are met by the subjects in this study. Owner/operators of all sizes of small firms (range 2 to 100; average number of employees 46) are included. All subjects hold the same organizational position, and all subjects have direct access to the information requested.

The second problem, measuring the effects of this additional boundary spanning activity, is dealt with in the procedures section.

Measures

The data reported here are part of a larger study, and only the relevant portions are included. Boundary spanning was measured in two ways.

Subjects were asked to record the number of hours and the number of contacts they had in an average week with each of nine categories of environmental constituencies. The categories were: (1) customers/clients; (2) suppliers/vendors; (3) interviewing new employees; (4) bankers, lawyers, accountants; (5) business, professional, and trade associations; (6) regulators, government, and union officials; (7) outside stockholders or creditors; (8) competitors; and (9) other business related activity. The sum of the hours spent in these activities divided by the number of hours worked per week is a measure of how intensive the owner/operator's boundary spanning activity is: the percentage of work time spent in boundary spanning activity. The sum of the number of contacts is an extensive measure. (The calculation of the variables used in the statistical tests reported below did not include customer/client figures because many subjects reported very large customer/client counts, indicating a cashier's role and not actually boundary spanning.)

Two measures of information processing capability were used. The first was intolerance of ambiguity (Budner, 1962). This is a 16-item questionnaire. Budner defined intolerance of ambiguity as the tendency to perceive ambiguous situations as sources of threat. The situations may be new, complex, or insoluble. As such, this definition represents the "content" dimension of information capability. Budner reported a series of reliabilities (Cronbach's alpha) ranging from .39 to .62 for the scale and admitted that the low reliability was because of the multidimensionality of the variable. The reliability for the subjects in this study was $\alpha = .56$.

The second measure used was integrative complexity, which was operationalized by the Schroder sentence completion test (Schroder et al., 1967). The three sentence items to which the subjects responded were:

"When I am in doubt...."

"Rules...."

"When I am criticized...."

Schroder et al. (1967) provide a scoring manual, and responses are scored along a 7-point Likert scale. The criterion for scoring is the complexity with which the answers are structured. The two highest scores are averaged for the final score. This method produces the same internal reliability as the correlation between total score and each item of a 6-item measure (Schroder et al., 1967). Cronbach's alpha for the two items in this study was .87.

Performance was measured by net addition to retained earnings, current sales, and the gross economic benefits accruing to ownership for 1981. This last measure was constructed by prorating by percentage of ownership the owner's draw, fringe benefits, cash dividends, and net additions to retained earnings. It represents the total economic benefits available to ownership (adjusted for more than one equity position). This measure was suggested by Webster (1977) and Susbauer (1978).

Testing the Hypotheses

Hypotheses 1 and 2 were tested using simple correlation coefficients. However, in Hypotheses 3 through 7, financial performance is the dependent

variable, and a number of influences on a firm's performance, other than the owner's boundary spanning and information capability, need to be considered. Data were collected on four of these possible influences: the type of business (a nominal variable), the age of the firm, the number of employees (size), and the total number of hours that all organizational members engaged in boundary spanning (as reported by the owner).

The variation in performance attributable to these four contextual influences was controlled for by entering these variables into the regressions on financial performance first, and then entering the independent variable of interest. The test used was the significance of the *F*-ratio due to the change in the R^2 for the regression (Kerlinger & Pedhazur, 1973).

Results

Evidence of Validity

An examination of Table 1 shows that there is evidence of construct validity for the general model of Figure 1. The correlations among the variables representing the three constructs—information processing capability, boundary spanning activity, and financial performance—all (with one exception) possess the correct sign and are statistically significant. The exception is the correlation between net additions to retained earnings and gross economic benefits. The near-zero correlation may indicate that in the short run, on an annual basis, the level of earnings attained does not affect executive compensation. Further evidence for this is found in the positive correlation between gross economic benefits and a 3-year average of earnings ($r = .35, p < .05$). This indicates that over a longer period executives' compensation is related to the level of earnings.

Another validity check concerns the assumption that the owner/operator is the key boundary spanner in the organization. This also is borne out, by both qualitative and quantitative data. The results show that for all boundary spanning activity (including customers), the owner/operator spends 33 working hours (59 percent of work time) and has 137 boundary spanning contacts per week. Even excluding customers, the owner spends over a quarter of his time (27 percent) engaged in boundary spanning.

Compare these figures to the amount of total organizational boundary spanning reported by the owner (in Table 1). Given that the owner's estimate is biased but that the bias is standardized, the owners report an average of 78 hours of boundary spanning for the entire firm. Thus, the owner's proportion of all the firm's boundary spanning is $(33/33 + 78)$ 29.7 percent. Further, if the bias is not standardized across firms, it most likely is represented in customer-related boundary spanning because this is the largest single category of boundary spanning reported. When customer-related boundary spanning is removed from the totals, the owners report an average of 15 hours of boundary spanning versus 12 hours for all other

Table 1
Means, Standard Deviations and Zero-Order Correlations
(Number of Cases in Upper Triangle)

	Mean	S.D.	1	2	3	4	5	6	7
1. Boundary spanning (pct. time)	26.8	22.6	—	68	74	66	59	54	52
2. Boundary spanning contacts	24.3	23.2	.38***	—	70	60	55	50	65
3. Intolerance of ambiguity	60.0	10.5	-.10	-.11	—	67	61	56	54
4. Integrative complexity	5.7	1.9	.21*	.02	.31**	—	55	51	49
5. Sales (\$)	1,795,525.	2,690,688.	.43***	.27*	-.29*	.06	—	55	53
6. Retained earnings (\$)	22,427.	37,069.	.12	.29*	.12	.01	.44***	—	55
7. Gross economic benefits (\$)	168,445.	430,797.	.61***	.06	-.16	-.04	.40***	.01	—
Contextual variables									
Total organization BSA (hours)	78.2	105.1	.09	.06	-.18	.20	.33**	.06	.24*
Number of employees	46.3	74.3	.08	.09	-.02	.19	.39**	.48***	.07
Age of firm (years)	29.5	22.0	.01	-.05	-.03	.12	-.04	-.03	.02
Type of business (nominal variable) ^a			.04	.16	.01	.04	.09	.04	.07

^aCorrelation reported is η^2 (proportion of variance explained).

* $p < .05$

** $p < .01$

*** $p < .001$

organizational participants. Thus, the owners report that they perform (15/12 + 15) 55.5 percent of all noncustomer boundary spanning themselves.

This information seems plausible when compared with qualitative information obtained from the informants. The owners were asked to list the job titles of employees who also engaged in boundary spanning activity. Of the 82 subjects, 29 reported no other boundary spanners. Only 18 reported purchasing agents or buyers. In fact, only six firms reported job titles that suggested boundary spanning of a strategic nature (three vice-presidents, two engineers, and one consultant). Taken together, the data strongly suggest that the owner/operator clearly is the dominant boundary spanner in the organization and that boundary spanning activity comprises a major entrepreneurial role.

Results of Hypothesis Tests

Table 1 shows the intercorrelations among the variables. The relationship between integrative complexity and boundary spanning as a percentage of time is positive and significant ($p < .05$). The relationship between intolerance of ambiguity and both measures of boundary spanning, though not statistically significant, is negative, and this is the expected sign. Therefore, Hypothesis 1 is accepted, and there is evidence in support of Hypothesis 2.

Hypotheses 3, 4, and 5, describe the effect of integrative complexity, intolerance of ambiguity, and boundary spanning activity on the financial performance of the firm. These hypotheses were tested by examining the incremental variance explained by the independent variable after the effects of the contextual variables were removed. Table 2 shows the results of the regressions of the contextual variables on financial performance.

Table 2
Results of Regression of the Contextual
Variables on Financial Performance

<i>Financial Performance</i>	<i>Variable</i>	<i>B</i>	<i>S.E. B</i>	<i>F</i>
Sales	Total organization BSA	12,029.	2,964.	16.5
	Age of firm	-241,250.	238,820.	1.0
	Type of business	-721,030.	385,888.	3.5
	Number of employees	20,240.	4,215.	23.0
	Constant	555,248.		
	$F_{4,50} = 8.54$ ($p < .001$)			
$R^2 = .41$ S.E. = 2,155,098	Total organization BSA	67.	45.	2.2
Retained earnings	Age of firm	-2,773.	3,698.	.6
	Type of business	-9,633.	5,976.	2.6
	Number of employees	285.	65.	19.1
	Constant	11,191.		
	$F_{4,46} = 5.5$ ($p < .001$)			
$R^2 = .31$ S.E. = 32,115	Total organization BSA	1,137.	630.	3.2
Gross economic benefits	Age of firm	3,596.	50,745.	.0
	Type of business	-8,408.	81,995.	.1
	Number of employees	846.	895.	.9
	Constant	30,313.		
	$F_{4,44} = .09$			
$R^2 = .08$ S.E. = 431,736				

Table 2 indicates that the contextual variables do explain a significant proportion of the variance in sales and retained earnings. In both of these cases, the number of employees (size) is the most important influence. However, these variables are not good predictors of gross economic benefits. It thus appears that this variable—which represents, in effect, executive compensation—is the most context free of the performance criteria.

Table 3 displays the results of the regressions of the information processing variables on financial performance. After the influence of the contextual variables was removed, neither intolerance of ambiguity nor integrative complexity made a significant contribution to the explained variance in financial performance. Therefore, Hypotheses 3 and 4 are rejected.

Table 4 shows the results of the test of Hypothesis 5. The upper portion of the table indicates that boundary spanning as a percentage of time contributes significantly and positively to the variance in financial performance for sales and gross. The lower portion of the table indicates that the number

Table 3
Effects of Information Processing
Capability on Financial Performance

<i>Financial Performance (N)</i>	<i>R² Context Variables</i>	<i>R² with Information Processing</i>	<i>Sign of Beta Coefficient</i>	<i>F-Ratio of R² Increment</i>
<i>Intolerance of Ambiguity</i>				
Sales (54)	.41***	.44***	—	2.57
Retained earnings (50)	.31***	.35***	+	2.70
Gross (48)	.08	.09	—	.46
<i>Integrative Complexity</i>				
Sales (54)	.41***	.41***	—	.00
Retained earnings (50)	.31***	.32***	—	.68
Gross (48)	.08	.10	—	.93

* $p < .10$

** $p < .05$

*** $p < .01$

Table 4
Effects of Boundary Spanning Activity
on Financial Performance

<i>Financial Performance (N)</i>	<i>R² Context Variables</i>	<i>R² with Boundary Spanning</i>	<i>Sign of Beta Coefficient</i>	<i>F-Ratio of R² Increment</i>
<i>Boundary Spanning as a Percentage of Time</i>				
Sales (54)	.41***	.53***	+	8.17***
Retained earnings (50)	.31***	.31**		.00
Gross (48)	.08	.42***	+	24.62***
<i>Boundary Spanning Contacts</i>				
Sales (54)	.41***	.43***	+	1.68
Retained earnings (49)	.31***	.35***	+	2.65
Gross (48)	.08	.08		.00

* $p < .10$

** $p < .05$

*** $p < .01$

of boundary spanning contacts is a positive influence on sales and retained earnings but not at statistically significant levels. The conclusion is that boundary spanning activity is positively related to financial performance, and the hypothesis is accepted.

Hypotheses 6 and 7 posited a stronger relationship between financial performance and boundary spanning when the owner/operator possessed higher levels of information processing capability. These hypotheses were tested in two complementary procedures (Argote, 1982). The first method used a split sample technique (disjoint analysis). The sample was divided at the median values for intolerance of ambiguity and integrative complexity. The effects of the contextual factors were removed, and the partial correlation coefficients between boundary spanning and financial performance were calculated. These partial correlation coefficients then were compared for the high and low groups. The results are presented in Table 5.

The results shown in Table 5 are mixed. The relationship between boundary spanning contacts and financial performance (Table 5, II, IV) appear substantially unchanged by information processing capability. If a pattern

Table 5
Correlations Between Boundary Spanning Activity and
Financial Performance for High and Low
Information Processing Capability^a

<i>Financial Performance (N_H, N_L)</i>	<i>Intolerance of Ambiguity</i>		<i>Z^b</i>
	<i>High</i>	<i>Low</i>	
<i>I. Boundary Spanning as a Percentage of Time</i>			
Sales (26, 27)	.36**	.47***	.46
Retained earnings (26, 23)	.31	-.31	-2.10**
Gross (24, 23)	.08	.70***	2.52***
<i>II. Boundary Spanning Contacts</i>			
Sales (24, 27)	.20	.18	-.07
Retained earnings (24, 23)	.44***	.08	-1.25
Gross (23, 23)	.33	-.05	-1.24
<i>Financial Performance (N_H, N_L)</i>	<i>Integrative Complexity</i>		<i>Z</i>
	<i>High</i>	<i>Low</i>	
<i>III. Boundary Spanning as a Percentage of Time</i>			
Sales (29, 24)	.43***	.44***	-.05
Retained earnings (26, 23)	-.16	.38**	-1.84**
Gross (25, 22)	.70***	.40**	1.42*
<i>IV. Boundary Spanning Contacts</i>			
Sales (29, 23)	.15	.36	-.76
Retained earnings (26, 21)	.31	.10	.70
Gross (25, 20)	.00	.07	-.22

^aThe correlations presented are the partial correlation coefficients between boundary spanning and financial performance after the variance due to the contextual variables has been removed.

^bPositive Z scores indicate that the correlations between boundary spanning and financial performance were greater under conditions of high information processing capability. Conversely, negative Z scores indicate that the correlations were greater under conditions of low information processing capability.

* $p < .10$

** $p < .05$

*** $p < .01$

emerges at all it is that the stronger boundary spanning contacts-performance relationships exist under low information processing capability (five of six signs are negative).

When boundary spanning as a percentage of time (intensive boundary spanning) is examined, significant relationships do emerge, but these are not consistent (Table 5, I, III). Low information processing capability seems to strengthen the boundary spanning-retained earnings relationship. High information processing capability seems to strengthen the boundary spanning-gross relationship. By itself, this test is inconclusive.

Table 6 presents the results of a moderated regression designed to test the incremental effects of the interaction of information processing capability and boundary spanning on financial performance. The contextual variables and main effects were regressed on performance (the additive model), and then the multiplicative interaction term was included in the regression (Kerlinger & Pedhazur, 1973).

This complementary test indicates that the strongest interaction occurs for boundary spanning as a percentage of time and gross. As before, information processing capability has little effect on the boundary spanning contacts-performance relationship. However, the inconsistent and unpredicted relationship between retained earnings and low information processing capability that emerged in the disjointive tests of Table 5 disappears in the conjointive tests of Table 6. The conclusion is that information processing capability does affect the boundary spanning (intensive type)-performance relationship (gross), and Hypotheses 6 and 7 are accepted.

Table 6
Effects of Information Processing Capability on Relationships Between Boundary Spanning Activity and Financial Performance

<i>Financial Performance (N)</i>	<i>R² Main Effects^a</i>	<i>R² Main and Interaction Effects</i>	<i>F-Ratio of R² Increment</i>
<i>I. Intolerance of Ambiguity and Boundary Spanning as a Percentage of Time</i>			
Sales (54)	.55***	.55***	.00
Retained earnings (50)	.34***	.35***	.00
Gross (48)	.42***	.63***	22.70***
<i>II. Intolerance of Ambiguity and Boundary Spanning Contacts</i>			
Sales (54)	.46***	.46***	.00
Retained earnings (49)	.39***	.44***	3.66*
Gross (48)	.09	.09	.00
<i>III. Integrative Complexity and Boundary Spanning as a Percentage of Time</i>			
Sales (54)	.55***	.58***	3.28*
Retained earnings (50)	.33***	.33***	.00
Gross (48)	.47***	.57***	9.30***
<i>IV. Integrative Complexity and Boundary Spanning Contacts</i>			
Sales (54)	.44**	.44**	.00
Retained earnings (49)	.36***	.38***	1.32
Gross (48)	.10	.10	.00

^aMain effects include contextual variables as well as boundary spanning and information processing capability.

* $p < .10$

** $p < .05$

*** $p < .01$

Discussion

The results show that the strategic management process and environment/organization interaction is alive and well even in the microbusiness. The amount of boundary spanning performed by the owner/operator indicates that the strategic alignment and negotiation process is a continuing concern for the small business owner even though the goal structure and strategic plans are implicit. More importantly, this strategic action is significantly related to the performance of the firm and the compensation of the entrepreneur.

Of course, not all boundary spanning is of a strategic nature. Although the entrepreneur spends a considerable amount of time and energy performing boundary role functions, some of this activity is directed at sales, negotiating terms with suppliers, and other day-to-day operations. But even the pattern of these more prosaic activities is the consequence of strategic decisions: targeting a market or a customer need, securing sources of raw materials and supplies. This indicates that the implementation of the strategic plan (however implicit) is performed directly by the entrepreneur.

The implication for current and future small business owner/operators is that the role of strategic planner must not be neglected. Regardless of the criteria for performance, the owner/operator must be sensitive to constituencies in the environment. There clearly is a need to perform environmental analysis and personally engage those elements of the environment that provide opportunities or pose risks for the firm.

The second set of findings concerns the difference in the types of boundary spanning that may be engaged in. Katz and Kahn (1978) suggested that both intensity (amount of time spent in boundary spanning) and extensity (number of contacts) are important measures. However, it appears that although the measures are theoretically and empirically related, they represent different types of behavior and serve different functions.

First, the intensive type of boundary spanning activity (percentage of time) is positively and significantly related to two of three measures of financial performance. Between 12 percent and 34 percent of the increase in explained variance in performance can be attributed to intensive boundary spanning activity. Thus it appears that although intensive boundary spanning is related to the number of contacts ($r = .38$, $p < .001$), it is a more performance related activity. It would appear that the owner/operator, having evaluated the important constraints and contingencies imposed by the environment, concentrates energy on selected environmental elements that are perceived to contribute most to performance. These may be different for any individual owner/operator, but the processes in environmental analysis, problem identification, risk and opportunity evaluation, and resource commitment seem to take place. This would mean that some form of strategic management or planning can be imputed from the owner/operator's boundary spanning activity.

A second difference between the two types of boundary spanning is found in the effects of information processing capability on the boundary spanning-performance relationship. For the most part, information processing capability had no effect on the relationship between the number of boundary spanning contacts and financial performance. However, the information processing characteristic appears to be an important talent for the owner/operator when engaged in intensive boundary spanning (and thereby strategic action). This is especially relevant when the performance criterion is the personal economic benefits of the owner/operator (gross). The boundary spanning activity in the high integrative complexity group and the low intolerance of ambiguity group is strongly related to personal economic gain. When lower levels of information processing capability are present, the intensive boundary spanning-performance relationship declines.

There also are implications for the development, selection, and training of individuals who desire to become owner/operators. For individuals with greater information processing capacity and complexity, skills that increase competency for intensive boundary spanning could be sharpened. For the other individuals, nonboundary spanning, internally oriented skills may be most appropriate. The point here is that the individual with lower levels of information processing capability may contribute more to performance in certain routine boundary spanning situations (Kolchin, 1980).

The study found only weak support for the hypotheses concerning the effects of the information processing variables on boundary spanning and performance. The only significant zero-order correlations were between intensive boundary spanning and integrative complexity ($r = .21, p < .05$) and sale and intolerance of ambiguity ($r = -.29, p < .01$). However, when the effects of the contextual variables were included in the analysis, there was no statistically significant relationship between information processing capability and financial performance. Also, the low reliability of the intolerance of ambiguity instrument diminishes the interpretability of these findings. Therefore, it is necessary to look elsewhere for constructs that explain the variation in boundary spanning itself. Among the most commonly suggested have been the environment (Aldrich & Herker, 1977; Mindlin & Aldrich, 1975; Thompson, 1967), organizational structure (Leifer, 1975; Leifer & Delbecq, 1978), technology (Galbraith, 1977; Thompson, 1967), size (Child, 1973; Hall, 1967), or innovation (Rogers & Rogers, 1976; Tushman, 1977).

Summary

This research provides evidence of two important aspects of strategic management in small business organizations. The first conclusion is that entrepreneurial boundary spanning does exist in microbusinesses. It is clear that the entrepreneur is deeply involved in environmental action, an important aspect of strategic management.

The second conclusion is that this environmental action is strongly related to financial performance criteria. That is, boundary spanning in the strategic management of small firms provides the firm with an edge in the marketplace. This is the same conclusion that would arise from the study of large complex organizations.

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Marc J. Dollinger is Assistant Professor in the College of Business and Economics, University of Kentucky.

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